

Board gender diversity, quotas, and ESG disclosure: Global evidence

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Highlights

- Gender-diverse boards around world significantly increase ESG disclosure.
- This positive effect of gender diversity on ESG becomes stronger in countries with weaker stakeholder regimes.
- This positive effect of gender diversity on ESG becomes stronger in countries with weaker and information environments.
- The positive role of gender-diverse boards on ESG is more pronounced in countries with less developed stock markets.
- The positive role of gender-diverse boards on ESG is more pronounced in countries with less developed credit markets.
- ESG disclosure significantly increases following the enactment of gender quotas in a country.

Abstract

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sample of 48 countries from 2005 to 2019, we show a significantly positive relationship between gender-diverse boards and ESG disclosure. This positive effect of gender diversity becomes stronger in countries with weaker stakeholder regimes and information environments, suggesting that those countries can obtain greater benefits by appointing more female directors. Further, using a staggered difference-in-differences design, we document a significant increase in ESG disclosure following the enactment of gender quotas. Overall, our findings demonstrate why gender-diverse boards for corporate social actions are relevant in an international setting.

Keywords

Board gender diversity; ESG disclosure; Stakeholder regimes; Information environments; Quotas

JEL classification

G30; G15; G43M41; M14

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Introduction

To address growing concerns about gender equality in senior management positions of modern corporations, policymakers around the world have adopted a number of regulations to enforce greater female representation on corporate boards.¹ Recently, there has been a significant increase in research exploring the role of gender-diverse boards in value-adding corporate policies; for example, novel patents (Griffin, Li, & Xu, 2021), buying shares back (Evgeniou & Vermaelen, 2017), empire building via acquisition (e.g., Huang & Kisgen, 2013; Levi, Li, & Zhang, 2014), and engagement in corporate social responsibility (CSR) initiatives (e.g., McGuinness, Vieito, & Wang, 2017; Post, Rahman, & Rubow, 2011). Extending this line of research and taking advantage of the staggered initiation of gender quotas around the world as a natural experiment, this study investigates whether appointing more female directors and gender quota

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mostly country-specific and inconclusive. For instance, some studies document a positive effect (Arayssi, Jizi, & Tabaja, 2020; Velte, 2016; Wasiuzzaman & Wan Mohammad, 2020), while another study finds no significant association between female directors and ESG disclosure (Manita, Bruna, Dang, & Houanti, 2018). The evidenced relation in those studies may be not generalisable to all countries for two key reasons. First, corporate governance systems and regulations differ across countries, and the regulatory environment in a country particularly determines female representation on corporate boards and CSR initiatives. For example, Norway, which is a global leader in CSR activities, has adopted mandatory gender quotas since 2006. Under the enforcement of this legislation, all firms listed publicly require at least 40% female representation on their boards of directors over a two-year grace period (Matsa & Miller, 2013) or the firm may be delisted for failure to meet the quota threshold. Other leading ESG European countries, such as Spain, France, and Belgium, have passed similar quota regulations (Spain 2007; France 2011 and Belgium 2011). In comparison with other adopters of female quota legislation, Norway has the highest degree of gender parity on corporate boards (Adams & Funk, 2012). Second, institutional settings, such as political, legal, and information environments, differ significantly between countries (Eleswarapu & Venkataraman, 2006). Countries with civil law (e.g., France and Germany) are more concerned with stakeholders than countries with common law (e.g., Australia and Canada) (Ball, Kothari, & Robin, 2000; Carey & Simnett, 2006). Against this backdrop, the influence of gender-diverse boards and quotas on ESG disclosure around the world, as well as how it differs among various institutional settings, remains an empirical issue. Our research fills this gap in the literature.

Drawing on stakeholder, gender socialisation, and ethicality theories, as well as existing empirical evidence, which is typically based mostly on a single country, we posit that firms with a gender-diverse board of directors are more sensitive to accountability issues relating to all stakeholders than firms without a gender-diverse board of directors and thus encourage the integration of ESG policies and practices into a firm's strategy and daily operations. To find strong support for our prediction, our sample includes 8262 firms across 48 countries over the period from 2005 to 2019. To quantify board gender diversity, we

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baseline analysis documents a significantly positive relation between female directors and ESG disclosure.² Our result is economically significant. For instance, a one standard deviation increase in the percentage of female directors on a corporate board causes a significant increase of 4.95% in ESG disclosure relative to the sample mean. This evidence suggests that firms with gender-diverse boards disclose higher levels of ESG issues to stakeholders and the public in international markets.³ Despite a favourable picture of female directors, the extent to which the documented relation is present depends on the institutional settings across the economies. Significant variations in regulatory attention to social and environmental issues exist in the community because of differences in stakeholder orientation (Dhaliwal, Li, Tsang, & Yang, 2014). Thus, we take advantage of differences in such institutional frameworks to further explore the role of gender-diverse boards in ESG disclosure. Firms in countries with a stronger stakeholder regime and well-established information environments are likely to be more concerned about stakeholders and thus, disclose more non-financial information. Therefore, in countries with weaker stakeholder frameworks and opaque information environments, we expect gender diversity to have a greater impact on ESG disclosure, as firms in these settings are expected to benefit more from the governance of female directors. Using measures for stakeholder orientation and information environments as in existing studies (Dhaliwal, Radhakrishnan, Tsang, & Yang, 2012; Djankov, McLiesh, & Shleifer, 2007; La Porta, Lopez-de-Silanes, & Shleifer, 2006; La Porta, Lopez-de-Silanes, Shleifer, & Vishny, 1997; La Porta, Lopez-de-Silanes, Shleifer, & Vishny, 1998), we document the governance role of female directors in ESG disclosure to be more pronounced for firms from countries, where the stakeholder regimes are weaker and information environments are poorer. Our findings suggest that firms in countries characterised by weaker stakeholder orientation and information environments accumulate greater benefits by appointing more female directors. The additional tests show that the governance role of gender-diverse boards on ESG is more pronounced in countries with less developed stock markets and credit markets. In contrast, the role of lower middle-income countries keeps silence in the relation between female directors and ESG disclosure.⁴ These results suggest			

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evidence for two reasons. First, omitted variables or unobserved firm characteristics driving both board gender diversity and ESG disclosure simultaneously are likely to bias our main results (Adams & Ferreira, 2009). Second, female directors are more interested in joining firms where the levels of social responsibility are higher (Atif et al., 2021), resulting in a self-selection bias. To rule out these concerns, we use a quasi-natural experiment with the staggered initiation of gender quotas across eight countries during the period of 2005 to 2016 and construct a large sample of firms with ESG disclosure information from 48 countries. Following Norway's introduction of a gender quota in 2003 (mandated in 2006), other countries implemented legislation imposing gender quotas on corporate boards.⁵ Using a staggered difference-in-differences (DID) design, we show that firms in countries that enacted gender quotas (quota countries) experience a significant increase in ESG disclosure in the post-law periods relative to firms from countries that never passed such laws (non-quota countries). To confirm our DID method on the influence of imposing gender quotas on ESG disclosure, we undertake many validation experiments, including a parallel trends assumption test, a confounding channel test, and a restricted sample test (i.e., only for quota-enacting countries).

To further control for endogeneity problems, we adopt several other methods to confirm that the positive effect of gender-diverse boards on ESG disclosure remains solid. First, we consider a firm fixed effects specification to account for time-invariant firm-specific omitted variables. Second, we use a propensity-score matching procedure, where we identify control firms without female directors that are otherwise indistinguishable from sample firms with female directors. Third, we employ a change analysis, as this approach provides a stronger test of causal relations than do levels of the variables due to year-to-year changes (Chung, Elder, & Kim, 2010). As a final check of endogeneity, we interact gender-diverse boards with variations originating from the legislation of gender quotas, given that enacting gender quotas is the best exogenous shock to board gender diversity. Overall, all these additional tests suggest consistent results, confirming the causal effect of gender diversity on ESG disclosure.

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Minguez-Vera, 2008; Carter, Simkins, & Simpson, 2003; Cumming, Leung, & Rui, 2015; Evgeniou & Vermaelen, 2017; Griffin et al., 2021; Huang & Kisgen, 2013; Liu, 2018; Liu & Wu, 2023; McGuinness et al., 2017; Post et al., 2011). In particular, Byron and Post (2016) in a meta-analysis of 87 independent studies conducted in different countries or multiple countries, reported a positive correlation between female board representation and CSR. Other studies have documented positive effects in both developed economies like Germany and Austria (Velte, 2016), Gulf Cooperation Council (GCC) countries (Arayssi et al., 2020), and developing economies like Malaysia (Wasiuzzaman & Wan Mohammad, 2020). Building on this line of research, we further validate the causal effect of gender-diverse boards on ESG and address the unsolved endogeneity problems in prior studies by using a staggered DID specification based on the enactment of gender quotas around the world. We find a significant increase in ESG disclosure in the post-law periods. Our results contribute to the ongoing debate on the valuation effects of gender-quota laws worldwide (e.g., Eckbo, Nygaard, & Thorburn, 2022; Ferrari, Ferraro, Profeta, & Pronzato, 2022). Second, we deepen the understanding of governance role of female board representation on ESG by addressing to what extent female directors can influence boards of directors and to what extent firms are motivated to fully utilize female directors' inclination to disclose more ESG matters in a global setting. Byron and Post (2016) suggest that the effectiveness of female directors in fostering ESG disclosure is contingent upon various institutional factors. Specifically, we test the extent to which a country's stakeholder protection, information environment, and level of development explain the influence of gender diversity on ESG disclosure. The main findings indicate that the governance role of gender-diverse boards is stronger (weaker) in those countries with weaker (better) market institutions in terms of stakeholder protection, information environment and market development. These results have important implication for promoting gender-diverse boards in the less developed economies and initiatives in different economies. Third, we add to the increasing amount of research about ESG disclosure determinants. In explaining ESG disclosure, recent literature mostly provides country-specific evidence; for example,			

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Rodionova, 2015). Those studies are limited to examining how a single country's institutional environment shapes corporate ESG. Lopatta, Jaeschke, and Chen (2017), using data from 25 countries, demonstrate the role of stakeholder engagement in the form of state-controlled ownership on the CSR performance of firms. Baldini, Dal Maso, Liberatore, Mazzi, and Terzani (2018), employing a cross-country sample during the period from 2005 to 2012, show evidence of how the labour system, the political system, and the system of culture have a significant impact on firms' ESG disclosure practices. Jackson, Bartosch, Avetisyan, Kinderman, and Knudsen (2020) document the role of mandatory non-financial disclosure regulation across 24 OECD countries in shaping CSR practices. However, in the international setting, there is a lack of evidence on the links between gender diversity in board members, ESG, and regulations. Extending this strand of literature, our study considers variations in the governance and legal and economic environments of different countries worldwide. Our research offers valuable explanation for ESG disclosure from a global viewpoint of stakeholder regimes, and information environments and economic incentives.

The rest of the paper is structured as follows. In Sections 2 and 3, we present an outline on the theoretical background, hypotheses development, as well as the research method, respectively. In Section 4, we discuss the role of gender diversity, institutional settings, and gender quotas on ESG disclosure, as well as other endogeneity designs. In Section 5, we present and conclude robustness checks, while we provide a summary of our paper in Section 6.

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Board gender diversity and ESG disclosure

The literature on board gender diversity is replete with evidence concerning the effect of female representation on boards on business decision-making (e.g., Adams, 2016; Atif et al., 2021). Various theories help to examine the relation between female directors and ESG disclosure. However, most prior studies focus on the female director's role from the perspectives of stakeholders.

Stakeholder theory, which stresses the interconnected relationships between a business and its varied ...

Data and sample

To study the link between board gender diversity, gender quotas, and ESG disclosure, we obtain data from several sources. To construct measures for board gender diversity, ESG disclosure, and firm-level accounting variables, we use the Bloomberg and Osiris databases. To quantify measures for economic development and institutional quality of a country, we use the World Development Indicators (WDI) and the World Governance Indicators (WGI), respectively. To gather information on the national ...

Board gender diversity and ESG disclosure: Baseline results

In Table 4, we report the regression results of ESG disclosure for H1 (Eq. (1)). In Panel A, we estimate ordinary least squares (OLS) as our base and regress ESG measures (ESG_t), including individual pillars (ENV_t , SOC_t , and GOV_t), on gender diversity ($FOBP_{t-1}$), controls, and various fixed effects (i.e., country, industry, and year). The coefficients of $FOBP_{t-1}$ across all the models (Columns 1–4) are all positive and significant at the 1% level, suggesting that the women on the board play a ...

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disclosure, we conduct a series of robustness tests. First, we explore whether there is a non-linear association between the number of women sitting on a firm's board of directors and ESG disclosure. Drawing from critical mass theory, when there is only one or two women on the board, then women board members do not have greater opportunities to influence the decision-making in modern corporations (Liu, 2018), suggesting a certain ...

Conclusion

Given the growing investor pressure toward sustainability leading to a boom in firms' information disclosure on ESG practices (Li et al., 2018) and a global call for gender diversity on boards of directors (Cumming et al., 2015), we examine the role of gender-diverse boards and gender quotas in explaining ESG-related information in international settings.

Using a worldwide sample of 8262 firms listed across 48 countries from 2005 to 2019, we document a significantly positive relation between ...

Acknowledgments

We appreciate helpful comments and suggestions on our paper from Andreanne Tremblay, Caroline Groskopf, Marc Steffen Rapp, and other participants at the 37th (2021) International Conference of the French Finance Association Conference (AFFI), the First (2021) EAA (The European Accounting Association) Virtual Annual Congress, and the 2020 Griffith University Seminar Series (AFE). ...

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...The unique values of female directors can provide different perspectives for decision-making (Carlson, 1972). Female directors are more willing to abide by the rules and to maintain long-term relationships with stakeholders, so they are more committed to promoting ESG practices (Alkhawaja et al., 2023; Beji et al., 2021). Male directors generally have a risk-taking and aggressive decision-making style and pay more attention to corporate financial return....

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